The Lefschetz Thimble for the Simple Harmonic Oscillator

A teaching-style derivation with worked examples

This note explains, from first principles, why the real-time Gaussian path integral for the simple harmonic oscillator can be made convergent by deforming the integration contour into complex space. It is written as a compact mathematical textbook chapter, with the missing background filled in carefully, including small examples of eigenvalues, eigenvectors, determinants, Jacobians, and Gaussian integrals.

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1 What problem are we trying to solve?

For the one-dimensional simple harmonic oscillator,

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2,$$

we would like to understand the real-time propagator

$$K(x_f, T; x_i, 0) = \langle x_f | \, \mathrm{e}^{-\frac{\mathrm{i}}{\hbar}HT} \, | x_i \rangle \, .$$

In the path-integral picture this becomes, formally,

$$K(x_f, T; x_i, 0) = \int_{x(0)=x_i}^{x(T)=x_f} \mathcal{D}x \; \exp\!\left(\frac{\mathrm{i}}{\hbar} S[x]\right),$$

where the action is

$$S[x] = \int_0^T \left[\frac{m}{2}\dot{x}^2 - \frac{m\omega^2}{2}x^2\right] \mathrm{d}t.$$

The main difficulty is that the factor

$$\exp\!\left(\frac{\mathrm{i}}{\hbar} S[x]\right)$$

is oscillatory rather than decaying. So instead of summing positive weights, we are summing complex arrows that point in many different directions. This is the sign or phase-cancellation problem.

The key idea of the Lefschetz thimble method is this,

> *Do not integrate over the original real variables if that contour is badly behaved. Deform the contour into complex space to a new manifold on which the phase is constant and the magnitude decays.*

For the harmonic oscillator this can be worked out exactly, which is why it is such a good teaching example.

2 Warm-up, one oscillatory Gaussian

Before touching the full path integral, it helps to understand the toy integral

$$I(\lambda) = \int_{-\infty}^{\infty} \mathrm{e}^{\frac{\mathrm{i}\lambda}{2}x^2} \mathrm{d}x, \qquad \lambda > 0.$$

On the real axis, the magnitude is

$$\left|\mathrm{e}^{\frac{\mathrm{i}\lambda}{2}x^2}\right| = 1,$$

so there is no decay at all. The integrand only rotates in the complex plane.

2.1 Why rotate by forty-five degrees?

Set

$$x = \mathrm{e}^{\mathrm{i}\pi/4} z, \qquad z \in \mathbb{R}.$$

Then

$$x^2 = \mathrm{e}^{\mathrm{i}\pi/2} z^2 = \mathrm{i}z^2,$$

so the exponent becomes

$$\frac{\mathrm{i}\lambda}{2} x^2 = \frac{\mathrm{i}\lambda}{2}(\mathrm{i}z^2) = -\frac{\lambda}{2} z^2.$$

Therefore,

$$\mathrm{e}^{\frac{\mathrm{i}\lambda}{2}x^2} \longrightarrow \mathrm{e}^{-\frac{\lambda}{2}z^2},$$

which is now a decaying Gaussian.

Also,

$$\mathrm{d}x = \mathrm{e}^{\mathrm{i}\pi/4} \mathrm{d}z,$$

so

$$I(\lambda) = \mathrm{e}^{\mathrm{i}\pi/4} \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{\lambda}{2}z^2} \mathrm{d}z.$$

Using the standard Gaussian formula

$$\int_{-\infty}^{\infty} \mathrm{e}^{-az^2} \mathrm{d}z = \sqrt{\frac{\pi}{a}}, \qquad a > 0,$$

we get

$$I(\lambda) = \mathrm{e}^{\mathrm{i}\pi/4} \sqrt{\frac{2\pi}{\lambda}}.$$

This is the simplest example of a thimble. The original real line has been replaced by a straight line in complex space making angle $\pi/4$ with the real axis.

2.2 A small numerical example

Take $\lambda = 2$. On the real axis,

$$\mathrm{e}^{\mathrm{i}x^2}$$

rotates rapidly as x grows.

At a few points,

x	x^2	$\mathrm{e}^{\mathrm{i}x^2}$
0	0	1
1	1	$\cos 1 + \mathrm{i}\sin 1$
2	4	$\cos 4 + \mathrm{i}\sin 4$
3	9	$\cos 9 + \mathrm{i}\sin 9$

But on the rotated contour $x = \mathrm{e}^{\mathrm{i}\pi/4} z$ we get

$$\mathrm{e}^{\mathrm{i}x^2} = \mathrm{e}^{-z^2},$$

which is positive and decays very quickly,

$$\mathrm{e}^{-0^2} = 1, \qquad \mathrm{e}^{-1^2} \approx 0.368, \qquad \mathrm{e}^{-2^2} \approx 0.018, \qquad \mathrm{e}^{-3^2} \approx 0.00012.$$

So the contour deformation turns a wildly oscillatory integral into a tame decaying one.

3 Discretising the harmonic-oscillator path integral

To make the path integral mathematically concrete, divide the interval $[0, T]$ into N equal pieces of size

$$\varepsilon = \frac{T}{N}.$$

Let the endpoints be fixed,

$$x_0 = x_i, \qquad x_N = x_f,$$

and treat the interior values

$$x_1, x_2, \ldots, x_{N-1}$$

as integration variables.

The discretised action is

$$S_N(\boldsymbol{x}) = \sum_{k=0}^{N-1} \Big[\frac{m}{2\varepsilon}(x_{k+1} - x_k)^2 - \frac{\varepsilon m \omega^2}{2} x_k^2 \Big],$$

where one often reorganises terms so that only the interior variables remain explicit.

For fixed endpoints, the integral becomes an $(N-1)$-dimensional oscillatory Gaussian,

$$K_N(x_f, T; x_i, 0) \propto \int_{\mathbb{R}^{N-1}} \exp\!\Big(\frac{\mathrm{i}}{\hbar} S_N(\boldsymbol{x}) \Big)\, \mathrm{d}^{N-1}x.$$

3.1 Quadratic form

Because the harmonic oscillator is quadratic, the action can be written as

$$S_N(\boldsymbol{x}) = \frac{1}{2} \boldsymbol{x}^{\mathsf{T}} A\, \boldsymbol{x} + \boldsymbol{b}^{\mathsf{T}} \boldsymbol{x} + c,$$

where A is a real symmetric tridiagonal matrix.

For the simplest case $x_i = x_f = 0$, the linear term vanishes and we get

$$S_N(\boldsymbol{x}) = \frac{1}{2} \boldsymbol{x}^{\mathsf{T}} A\, \boldsymbol{x}.$$

The matrix A has the form

$$A = \frac{m}{\varepsilon} \begin{pmatrix} 2-\varepsilon^2\omega^2 & -1 & 0 & \cdots & 0 \\ -1 & 2-\varepsilon^2\omega^2 & -1 & \cdots & 0 \\ 0 & -1 & 2-\varepsilon^2\omega^2 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & -1 \\ 0 & 0 & \cdots & -1 & 2-\varepsilon^2\omega^2 \end{pmatrix}.$$

So the whole real-time path integral is an ordinary finite-dimensional oscillatory Gaussian integral.

4 What are eigenvalues and eigenvectors, and why do we care?

Since the quadratic form is controlled by the matrix A, the natural way to simplify the integral is to diagonalise A.

4.1 Definition

For a square matrix M, an eigenvector is a non-zero vector $\boldsymbol{v}$ such that

$$M\boldsymbol{v} = \lambda\boldsymbol{v},$$

for some scalar λ. That scalar is the corresponding eigenvalue.

This means that applying M to $\boldsymbol{v}$ does not change its direction, only its size and possibly its sign or phase.

4.2 Tiny example

Take

$$M = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

Then

$$M\begin{pmatrix} 1 \\ 0 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad M\begin{pmatrix} 0 \\ 1 \end{pmatrix} = 3\begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

So the vectors $(1,0)^{\mathsf{T}}$ and $(0,1)^{\mathsf{T}}$ are eigenvectors, with eigenvalues 2 and 3.

4.3 Why diagonalising helps

If a symmetric matrix A can be written as

$$A = U\Lambda U^{\mathsf{T}},$$

where U is orthogonal and

$$\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_{N-1}),$$

then changing variables to the normal-mode coordinates

$$\boldsymbol{x} = U\boldsymbol{q}$$

gives

$$\boldsymbol{x}^{\mathsf{T}} A \boldsymbol{x} = \boldsymbol{q}^{\mathsf{T}} \Lambda \boldsymbol{q} = \sum_{n=1}^{N-1} \lambda_n q_n^2.$$

So instead of one coupled multidimensional oscillatory Gaussian, we get a product of one-dimensional oscillatory Gaussians.

That is exactly what we want.

5 Eigenvalues and eigenvectors of the discrete oscillator matrix

The tridiagonal matrix above is a standard discrete second-derivative operator plus a diagonal potential term. Its eigenvectors are discrete sine waves,

$$\left(v^{(n)}\right)_k = \sin\left(\frac{\pi n k}{N}\right), \qquad n = 1, 2, \ldots, N-1, \qquad k = 1, 2, \ldots, N-1.$$

The corresponding eigenvalues are

$$\lambda_n = \frac{m}{\varepsilon}\left(2 - 2\cos\frac{\pi n}{N} - \varepsilon^2\omega^2\right).$$

Using

$$2 - 2\cos\theta = 4\sin^2\frac{\theta}{2},$$

this can also be written as

$$\lambda_n = \frac{m}{\varepsilon}\left(4\sin^2\frac{\pi n}{2N} - \varepsilon^2\omega^2\right).$$

5.1 Why sine waves appear

The endpoints are fixed, so the allowed modes must vanish at both ends. Sine functions do exactly that,

$$\sin 0 = 0, \qquad \sin(n\pi) = 0.$$

So the discrete normal modes are just the lattice version of the familiar standing-wave modes.

5.2 Worked example, $N = 4$

Let $N = 4$, so there are $N - 1 = 3$ interior variables. Then the eigenvectors are

$$v^{(1)} = \begin{pmatrix} \sin(\pi/4) \\ \sin(\pi/2) \\ \sin(3\pi/4) \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} \\ 1 \\ \frac{\sqrt{2}}{2} \end{pmatrix},$$

$$v^{(2)} = \begin{pmatrix} \sin(\pi/2) \\ \sin(\pi) \\ \sin(3\pi/2) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix},$$

$$v^{(3)} = \begin{pmatrix} \sin(3\pi/4) \\ \sin(3\pi/2) \\ \sin(9\pi/4) \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} \\ -1 \\ \frac{\sqrt{2}}{2} \end{pmatrix}.$$

These are the discrete normal modes.

6 Reducing the path integral to independent modes

After diagonalisation, the action becomes

$$S_N = \frac{1}{2} \sum_{n=1}^{N-1} \lambda_n q_n^2.$$

So the path integral factorises into

$$K_N \propto \prod_{n=1}^{N-1} \int_{-\infty}^{\infty} \exp\left(\frac{\mathrm{i}}{2\hbar} \lambda_n q_n^2 \right) \mathrm{d}q_n.$$

Each mode is now exactly the one-dimensional oscillatory Gaussian from the warm-up section.

Therefore, for each mode with $\lambda_n > 0$, the natural thimble is

$$q_n = \mathrm{e}^{\mathrm{i}\pi/4} z_n, \qquad z_n \in \mathbb{R}.$$

If one had a negative eigenvalue, the steepest-descent direction would instead involve $-\pi/4$. In the stable harmonic-oscillator setting with a suitable regulator, the relevant directions are chosen so the real part of the exponent is negative along the thimble.

7 The thimble in the full multidimensional space

In the diagonal basis, the thimble is just the product of one-dimensional rotated lines,

$$q_n = \mathrm{e}^{\mathrm{i}\pi/4} z_n, \qquad z_n \in \mathbb{R}.$$

Equivalently,

$$\boldsymbol{q} = \mathrm{e}^{\mathrm{i}\pi/4} \boldsymbol{z}, \qquad \boldsymbol{z} \in \mathbb{R}^{N-1}.$$

Transforming back to the original coordinates,

$$\boldsymbol{x} = U\boldsymbol{q} = U\mathrm{e}^{\mathrm{i}\pi/4} \boldsymbol{z}.$$

So the full thimble is the image of $\mathbb{R}^{N-1}$ under the complex linear map

$$\boldsymbol{z} \mapsto \boldsymbol{x} = A\boldsymbol{z}, \qquad A := U\mathrm{e}^{\mathrm{i}\pi/4}.$$

More precisely, when one also rescales each mode to absorb the size of the eigenvalue, a convenient parameterisation is

$$q_n = \mathrm{e}^{\mathrm{i}\pi/4}\frac{z_n}{\sqrt{\lambda_n/\hbar}},$$

so that

$$\frac{\mathrm{i}}{2\hbar}\lambda_n q_n^2 = -\frac{1}{2}z_n^2.$$

In matrix form,

$$\boldsymbol{x} = U\ \mathrm{diag}\left(\mathrm{e}^{\mathrm{i}\pi/4}\sqrt{\frac{\hbar}{\lambda_n}}\right)\boldsymbol{z}.$$

This is the exact linear thimble map for the Gaussian problem.

8 What is a determinant, and why does it appear?

Whenever we change variables in an integral, the volume element changes. In one dimension,

$$\mathrm{d}x = \frac{\mathrm{d}x}{\mathrm{d}z}\mathrm{d}z.$$

In many dimensions the analogue is the determinant of the Jacobian matrix.

8.1 Tiny determinant example

For

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

its determinant is

$$\det M = ad - bc.$$

Geometrically, this gives the signed area-scaling factor of the linear map.

Example,

$$M = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \qquad \Longrightarrow \qquad \det M = 6.$$

So a unit square becomes a rectangle of area 6.

8.2 Jacobian in the thimble map

If

$$\boldsymbol{x} = A\boldsymbol{z},$$

then

$$\mathrm{d}^{N-1}x = \det(A)\,\mathrm{d}^{N-1}z.$$

In our Gaussian thimble map,

$$A = U \operatorname{diag}\left(\mathrm{e}^{\mathrm{i}\pi/4}\sqrt{\frac{\hbar}{\lambda_n}}\right).$$

Since U is orthogonal, $\det(U) = \pm 1$, so up to a sign,

$$\det(A) = \prod_{n=1}^{N-1} \mathrm{e}^{\mathrm{i}\pi/4}\sqrt{\frac{\hbar}{\lambda_n}} = \mathrm{e}^{\mathrm{i}(N-1)\pi/4} \prod_{n=1}^{N-1} \sqrt{\frac{\hbar}{\lambda_n}}.$$

This determinant is exactly the multidimensional generalisation of the one-dimensional factor $\mathrm{e}^{\mathrm{i}\pi/4}$ from the toy Gaussian.

9 Putting the pieces together

After diagonalising and rotating each mode onto its thimble,

$$K_N \propto \prod_{n=1}^{N-1} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}z_n^2\right)\left(\mathrm{e}^{\mathrm{i}\pi/4}\sqrt{\frac{\hbar}{\lambda_n}}\right)\mathrm{d}z_n.$$

Since

$$\int_{-\infty}^{\infty} \mathrm{e}^{-z_n^2/2}\mathrm{d}z_n = \sqrt{2\pi},$$

we obtain

$$K_N \propto \mathrm{e}^{\mathrm{i}(N-1)\pi/4} \prod_{n=1}^{N-1} \sqrt{\frac{2\pi\hbar}{\lambda_n}}.$$

Equivalently,

$$K_N \propto \frac{\mathrm{e}^{\mathrm{i}(N-1)\pi/4}}{\sqrt{\det A_{\mathrm{quad}}}},$$

where A_{quad} here means the quadratic-form matrix of the discretised action, not the thimble map. The product of eigenvalues is the determinant,

$$\det A_{\mathrm{quad}} = \prod_{n=1}^{N-1} \lambda_n.$$

That is one reason eigenvalues are so useful, they turn determinants into products.

10 A fully worked small example

Let us take

$$N = 4, \qquad T = 1, \qquad m = 1, \qquad \omega = 1, \qquad \hbar = 1.$$

Then

$$\varepsilon = \frac{T}{N} = \frac{1}{4}.$$

The quadratic matrix is

$$A = \frac{1}{\varepsilon}\begin{pmatrix} 2-\varepsilon^2 & -1 & 0 \\ -1 & 2-\varepsilon^2 & -1 \\ 0 & -1 & 2-\varepsilon^2 \end{pmatrix} = 4\begin{pmatrix} \frac{31}{16} & -1 & 0 \\ -1 & \frac{31}{16} & -1 \\ 0 & -1 & \frac{31}{16} \end{pmatrix} = \begin{pmatrix} 7.75 & -4 & 0 \\ -4 & 7.75 & -4 \\ 0 & -4 & 7.75 \end{pmatrix}.$$

10.1 Eigenvalues

Using

$$\lambda_n = \frac{1}{\varepsilon}\left(2 - 2\cos\frac{\pi n}{4} - \varepsilon^2\right),$$

we get

$$\lambda_1 = 4\left(2 - \sqrt{2} - \frac{1}{16}\right) \approx 2.109,$$

$$\lambda_2 = 4\left(2 - 0 - \frac{1}{16}\right) = 7.75,$$

$$\lambda_3 = 4\left(2 + \sqrt{2} - \frac{1}{16}\right) \approx 13.391.$$

10.2 Mode-by-mode thimble scaling

For each mode,

$$q_n = \mathrm{e}^{\mathrm{i}\pi/4}\frac{z_n}{\sqrt{\lambda_n}}.$$

So numerically,

$$q_1 \approx \mathrm{e}^{\mathrm{i}\pi/4}(0.6887)z_1, \qquad q_2 \approx \mathrm{e}^{\mathrm{i}\pi/4}(0.3592)z_2, \qquad q_3 \approx \mathrm{e}^{\mathrm{i}\pi/4}(0.2733)z_3.$$

This shows something important,

> The thimble is not only a rotation. It is a rotation *plus a mode-dependent rescaling.* Softer modes, with smaller eigenvalue, are broader. Stiffer modes, with larger eigenvalue, are narrower.

10.3 Determinant check

The product of eigenvalues is

$$\lambda_1\lambda_2\lambda_3 \approx 2.109 \times 7.75 \times 13.391 \approx 218.75.$$

So

$$\sqrt{\det A} \approx \sqrt{218.75} \approx 14.790.$$

Hence the Gaussian prefactor contains roughly

$$\frac{\mathrm{e}^{3\mathrm{i}\pi/4}}{14.790}$$

up to the usual $(2\pi\hbar)^{(N-1)/2}$ factor and endpoint normalisation constants.

11 Why this is called a Lefschetz thimble

The general theory starts from a complexified action and studies the flow of

$$\mathrm{Re}\left(\frac{\mathrm{i}}{\hbar}S\right)$$

in complex space. A thimble is the manifold of steepest descent attached to a critical point of the action.

For a Gaussian integral,

$$S = \frac{1}{2}\lambda q^2,$$

the critical point is simply $q = 0$. The steepest-descent contour through that point is the line on which

$$\mathrm{Im}\left(\frac{\mathrm{i}}{\hbar}S\right)$$

is constant and

$$\mathrm{Re}\left(\frac{\mathrm{i}}{\hbar}S\right)$$

decreases away from the critical point.

That condition leads exactly to the $\pi/4$ rotation.

In the full harmonic oscillator, the multidimensional thimble is just the product of these mode-wise steepest-descent directions.

12 Connection to the exact continuum propagator

For the harmonic oscillator, the exact continuum propagator is known,

$$K(x_f, T; x_i, 0) = \sqrt{\frac{m\omega}{2\pi\mathrm{i}\hbar\sin(\omega T)}}\exp\left[\frac{\mathrm{i}m\omega}{2\hbar\sin(\omega T)}\Big((x_f^2 + x_i^2)\cos(\omega T) - 2x_i x_f\Big)\right].$$

The discrete thimble derivation above is the finite-dimensional version of the same Gaussian physics. As $N \to \infty$, the discrete normal modes approach the continuum standing-wave modes, and the determinant of the discrete quadratic form approaches the continuum fluctuation determinant.

So the thimble construction is not a different answer. It is a clean way of computing the same answer by moving to the contour where the oscillatory Gaussian becomes an ordinary decaying Gaussian.

13 What you should remember

1. The harmonic-oscillator path integral is Gaussian once time is discretised.
2. The quadratic form is controlled by a symmetric tridiagonal matrix.
3. Diagonalising that matrix gives independent normal modes.

4. Each mode is an oscillatory Gaussian of the form

$$\int \mathrm{e}^{\mathrm{i}\lambda q^2/2\hbar}\mathrm{d}q.$$

5. The thimble for one such mode is the forty-five degree contour

$$q = \mathrm{e}^{\mathrm{i}\pi/4}z.$$

6. In many dimensions, the full thimble is the product of these mode-wise rotations, together with the correct eigenvalue-dependent rescalings.

7. The Jacobian determinant records the volume scaling of the contour deformation.

8. The whole method works because on the thimble the exponent becomes negative real,

$$\frac{\mathrm{i}}{2\hbar}\lambda q^2 \to -\frac{1}{2}z^2,$$

so the integral becomes convergent and easy to evaluate.

14 A compact formula sheet

For the zero-endpoint discretised harmonic oscillator,

$$S_N(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^{\mathsf{T}}A\boldsymbol{x},$$

with

$$A = \frac{m}{\varepsilon}\begin{pmatrix} 2-\varepsilon^2\omega^2 & -1 & \cdots & 0 \\ -1 & 2-\varepsilon^2\omega^2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ 0 & \cdots & -1 & 2-\varepsilon^2\omega^2 \end{pmatrix}.$$

Its eigenvectors are discrete sines,

$$\left(v^{(n)}\right)_k = \sin\left(\frac{\pi nk}{N}\right),$$

and eigenvalues are

$$\lambda_n = \frac{m}{\varepsilon}\left(2 - 2\cos\frac{\pi n}{N} - \varepsilon^2\omega^2\right).$$

Mode variables satisfy

$$S_N = \frac{1}{2}\sum_n \lambda_n q_n^2.$$

The exact Gaussian thimble map is

$$q_n = \mathrm{e}^{\mathrm{i}\pi/4}\sqrt{\frac{\hbar}{\lambda_n}}\,z_n, \qquad z_n \in \mathbb{R},$$

so

$$\frac{\mathrm{i}}{2\hbar}\lambda_n q_n^2 = -\frac{1}{2}z_n^2.$$

The Jacobian is

$$\det J = \mathrm{e}^{\mathrm{i}(N-1)\pi/4}\prod_{n=1}^{N-1}\sqrt{\frac{\hbar}{\lambda_n}}.$$

Final remark

For the harmonic oscillator, the thimble is exactly solvable because the action is quadratic. That is why it is the right place to build intuition. In interacting theories, such as the anharmonic oscillator, the same geometric idea survives, but the contour is no longer a simple linear rotation. That is where numerical contour deformation, flow methods, and machine learning become genuinely useful.